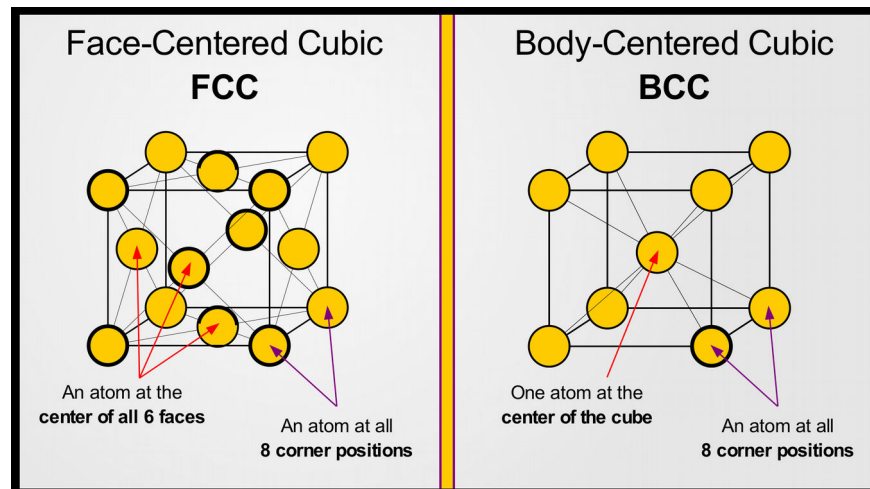


APS110 – Lecture 9

See hand written notes for drawings.

Body Centered Cubic

- Another metallic structure → means, only ONE type of atom
 - Rock Salt = ceramic structure, multiple types of atoms
- BCC has **contact of atoms along cube diagonals**



Coordination Number

→ Number of atoms that are contacting any given atom (nearest neighbor atoms)

- For BCC the coordination number would be the SAME for ANY given atom
 - $CN_{BCC} = 8$
 - Think of the central atom – contacting each of the corner atoms and there are 8 corners
- For FCC...
 - The right FC atom (on the face of the right side of the cube) is touching the **corner** atoms of the same face, but it is also touching the FC atoms from the **top** face, the **front** face, the **back** face, and the **bottom** face
 - $CN_{FCC} = 12$

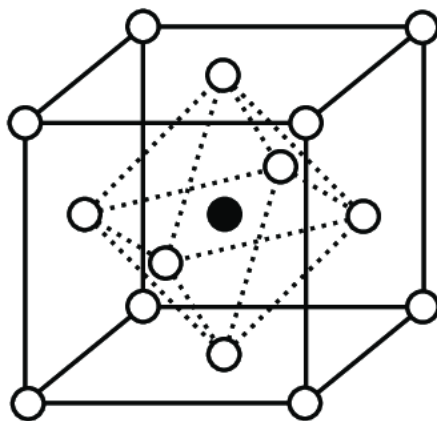
Fraction of volume occupied by atoms

- Volume packing fraction (or, atomic packing factor, APF)
- $APF = \text{volume of atoms} / \text{volume of unit cell}$
- $APF_{FCC} = n \cdot V_{\text{sphere}} / V_C = (4)(4/3 \pi \cdot R^3) / a^3$
 - $APF_{FCC} = 0.74$ ← makes intuitive sense (since most of the space of the cube is occupied by the volume of the spheres)
- $APF_{BCC} = 0.68$
 - Smaller than FCC – not as close packed
- FCC is a **close-packed** structure

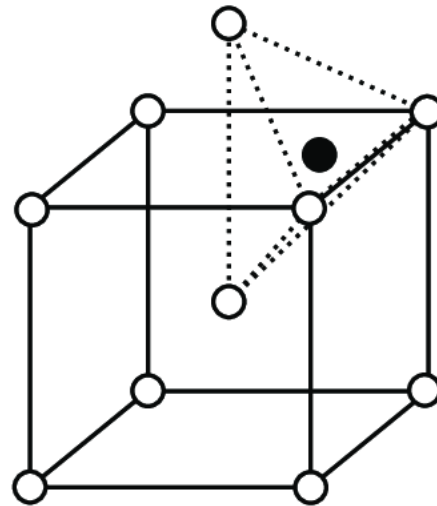
- o 0.74 is the smallest volume you can achieve packing spheres of the same diameter
- o FCC is sometimes called “cubic close packed”

Interstitial Sites

- The space between other atoms
- We’ve seen one already....in Rock Salt
 - o Look at the central atom (cation) – touches top, bottom, front, back, left, right
 - $CN = 6$ ← coordination number of cations in rock salt structure
 - o When we draw the connections for the central atom, the solid geometry forms a shape with 8 faces → **octahedral interstitial site**



fcc



bcc

- We now want to ask, how big is the interstitial site for that central cation?
 - o Essentially, we want to know how big of cation would fit in this space?
 - o This could help us learn how to predict crystal structure, given atomic radius